

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 9

Topology: Structure Without Measurement

Parallel Reading

Self-Study Curriculum

Parallel Reading: Jeffrey Weeks, *The Shape of Space* (2nd ed., 2002), Chapters 1-8 (approximately 100 pages). This is a visual, non-technical introduction to topology that builds intuition through examples: it is the main text this week.

Also read: PCM, sections III.90 (“Topological Spaces”) and III.9 (“Compactness and Compactification”): short reference entries, a few pages each. If you want the algebraic-invariant story behind the fundamental group, skim the opening pages of IV.6 (“Algebraic Topology”).